

Team Contribution Statement

UIB 11755 — Human-Computer Interaction, Final Project (WA4)

FoodLens: Transparent Multi-Objective Food Recommendations

All four team members contributed actively and equitably to both the research phase (WA3) and the implementation phase (WA4) of this project. The distribution of features was calibrated to balance total effort: Alejandro Bordón carried the largest share of academic writing across both assignments (WA3 paper, WA4 paper), so his implementation features were intentionally scoped to UI and localStorage work with no ML components (~40h). The remaining three members each took on broader implementation loads (~65–75h).

Alejandro Rafael Bordón Duarte (P1) — ~75h total (~40h implementation + ~35h academic writing) · 8 features

WA3 contributions (academic)

Authored the Introduction, Related Work, and Conclusion sections. Conducted the self-interview as participant P01. Led the overall document structure and coordination across WA3.

WA4 contributions (academic)

Authored the Introduction, Related Work, and Conclusion sections of the final paper. Led the overall document structure and coordination across WA4.

WA4 implementation — Batch H: Personal data layer (UI + localStorage, no ML)

- F-11** — Onboarding modal: first-visit goals/age/gender form, persists to localStorage.foodlens.profile *[H6]*
 - F-25** — Recently viewed history: last 10 products with thumbnails and dual-axis badges *[KI-5]*
 - F-26** — Favourites / Saved: heart toggle on every card, dedicated Saved view *[H7]*
 - F-27** — Settings page: unit system, language preference, default slider weight, clear profile *[KI-5]*
 - F-28** — Educational tooltips on Nutri-Score and Eco-Score badges, with links to official sources *[KI-1]*
 - F-29** — Share product: copies the Open Food Facts URL to clipboard, toast confirmation *[H7]*
 - F-30** — Print-friendly product card: CSS @media print rule, A4-ready layout *[H7]*
 - F-31** — Personas showcase block: static section above footer presenting Marc, Pau, and Lluís *[storytelling]*
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WA3 contributions (academic)

Authored the Methodology and Participants sections. Conducted the self-interview as participant P03. Designed and applied the semi-structured interview protocol.

WA4 contributions (academic)

Authored the Evaluation section of the final paper, including the SUS and Explanation Satisfaction Scale protocol design, pre-registration of hypotheses, and discussion of results.

WA4 implementation — Batch I: Comparison, filtering, i18n, and exports

F-18 — Compare with my usual: mark one product per category as reference; contrastive sentence updates accordingly *[H7 + KI-4]*

F-19 — Compare-this-shelf: add up to 5 products, ranked list with one-line justification per product *[H5]*

F-20 — Multi-criteria filter chips: high protein, low sodium, low CO₂, plastic-free, organic *[brief §3]*

F-22 — Category browser: hierarchical OFF category browser, top 10 pre-rendered, further levels lazy-loaded *[brief §3]*

F-32 — Allergen filter: chips for gluten, lactose, nuts, soy, egg, fish; persists per session *[KI-2]*

F-33 — Internationalisation: Spanish / English / Catalan via JSON dictionaries and a t(key) function *[localisation]*

F-34 — Multi-product comparison table: columns per product, rows per nutrient, winner highlighted per row *[KI-4]*

F-47 — CSV export of comparisons and favourites: pure client-side Blob, works from F-19, F-34, and F-26 *[KI-4]*

WA3 contributions (academic)

Authored the Data Analysis section, applying Braun and Clarke's six-phase thematic analysis framework to the six interview transcripts. Produced the codebook, affinity mapping, and saturation assessment.

WA4 contributions (academic)

Authored the XAI Integration section of the final paper, covering the contrastive sentence generator, visual feature attribution, and the layered explanation architecture (sentence → SHAP drill-down → scatter plot).

WA4 implementation — Batch J: External integrations, performance, accessibility, and WA5 forms

- F-21** — Barcode scanning via webcam using @zxing/browser; permission requested only on camera icon click *[H5]*
 - F-35** — Geolocation-based seasonal product hints: opt-in banner showing out-of-season import context *[KI-2]*
 - F-36** — Progressive Web App: manifest.json + service worker caching the static shell and sample data *[infra]*
 - F-37** — WCAG accessibility audit and fixes: focus rings, ARIA labels, contrast ratios, keyboard navigation *[a11y]*
 - F-38** — Performance pass: lazy image loading, Google Fonts subsetting, script deferral, Lighthouse ≥90 *[perf]*
 - F-39** — Dark mode “paper-night” theme via CSS [data-theme=dark] variables, toggled from settings (F-27) *[KI-5]*
 - F-40** — Search by ingredient using OFF /search?ingredients_tags=... *[brief §3]*
 - F-46** — In-app SUS (10 items, auto-scored) + Explanation Satisfaction Scale forms; exports results as JSON *[WA5]*
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WA3 contributions (academic)

Conducted the Open Food Facts pipeline technical feasibility exploration (Appendix D of the WA3 paper): OFF API v2 data acquisition, cleaning, Random Forest Nutri-Score classifier, KNN Alternative Engine, and DuckDB scaling check. Contributed to the persona synthesis and authored the Technical Feasibility appendix.

WA4 contributions (academic)

Authored the System Architecture and Implementation section of the final paper (Section 4), covering the data pipeline, the weighted KNN recommender with its distance function and strict-better filter, and the three-level asymmetric personalisation model.

WA4 implementation — Batch K: ML core, data visualisation, telemetry, and infrastructure

F-24 — SHAP-based local explanations: TreeExplainer over the RandomForest, waterfall + summary via Chart.js, behind Advanced toggle *[H2 advanced]*

F-41 — RandomForest retrained on the full OFF catalogue via DuckDB ($\geq 80\%$ CV accuracy); replaces the 60% on-the-fly demo *[ML]*

F-42 — Collaborative filtering layer on top of KNN: co-occurrence matrix blended with nutrient-space KNN; depends on F-45 telemetry data *[recomendación]*

F-43 — Nutri-Score $\times$ Eco-Score scatter plot per category: Pareto frontier visualisation; fed by F-41's full-catalogue model *[KI-3]*

F-44 — Redis cache layer replacing in-process TTLCache; Unicorn-ready, added as a service in docker-compose.yml *[infra]*

F-45 — Opt-in telemetry: logs decision time, alternative click-through, and slider changes; JSON-lines backend; upstream for F-42 *[WA5]*

Signed,

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Palma, June 2026